



Toegepaste Micro-economie: Public economics

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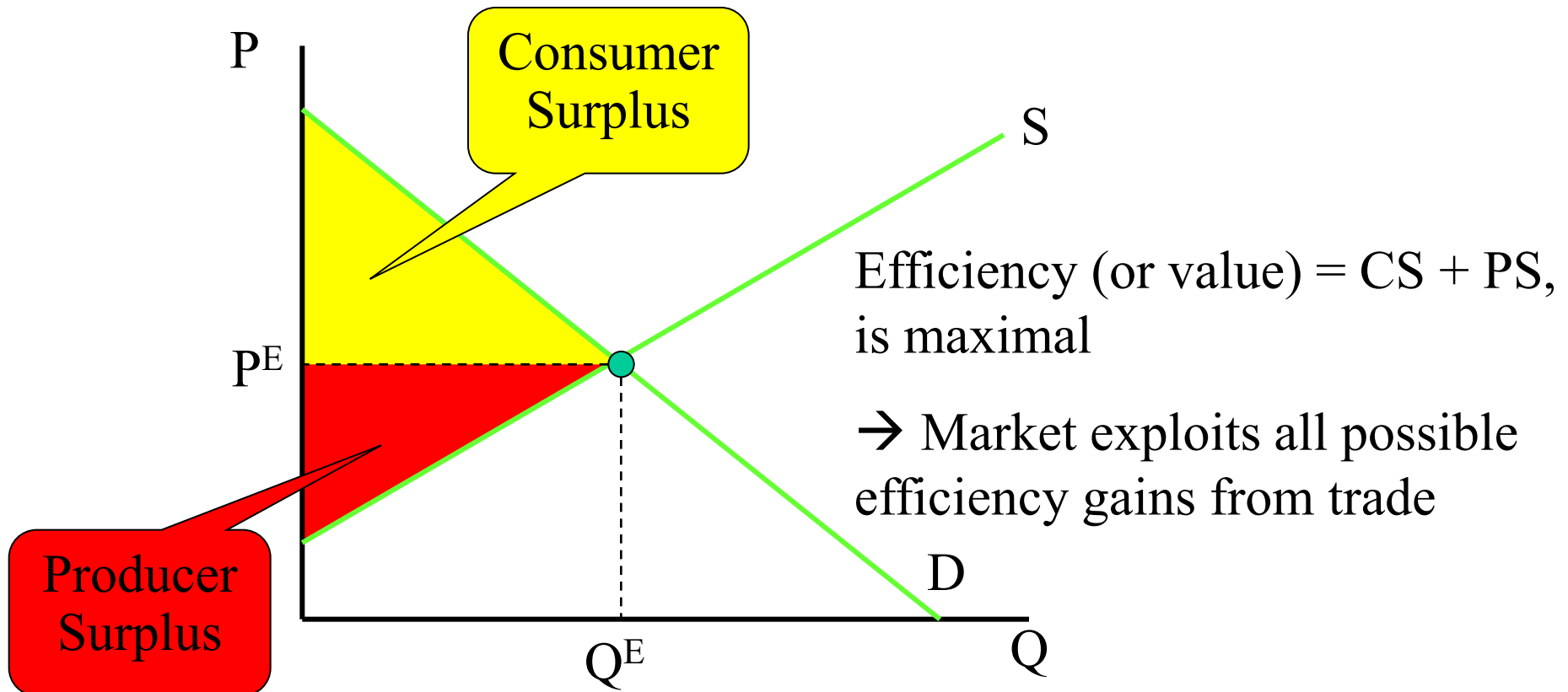
Lecture 1.5

- Review of Perfect Competition
 - Chapter 3 in R&G; (or Chapter 16 in Frank)
 - I assume: Individual consumer choice theory is known
 - Indifference curves and MRS; budget constraints
 - If not: Appendix R&G (or better: Chapter 3 in Frank)
- Key issues in trade-off between public vs private provision / production
 - Trade-offs in provision
 - Trade-offs in production
 - Valuation of intangibles in cost-benefit analysis

Perfect Competition

Market equilibrium

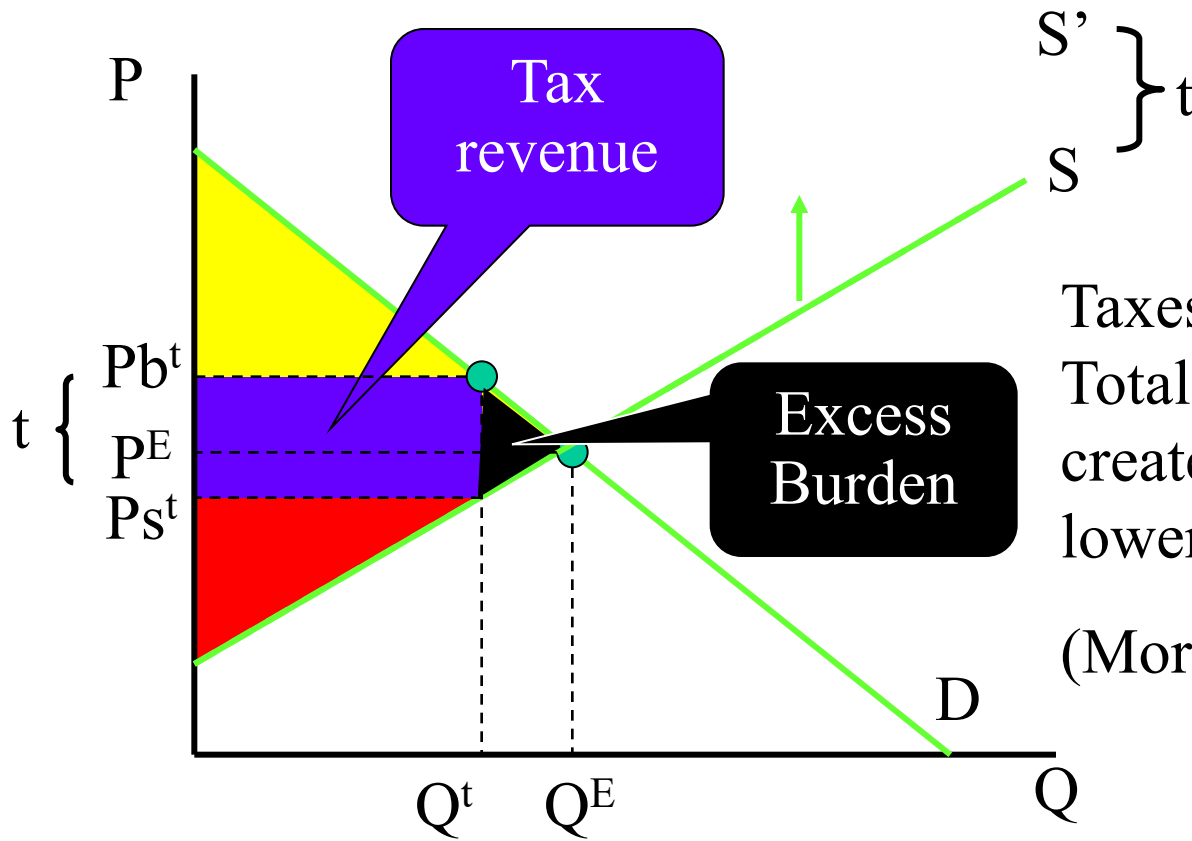
“Invisible hand” sets price such that Demand equals Supply



Perfect Competition

Market equilibrium with a unit tax t

Tax creates wedge between what buyers pay and what sellers get



Taxes are distortionary:
Total efficiency (value
created by the market) is
lower, by Excess Burden
(More on this in week 5)

Perfect Competition (video!)

Two good - two person exchange economy

Edgeworth box:

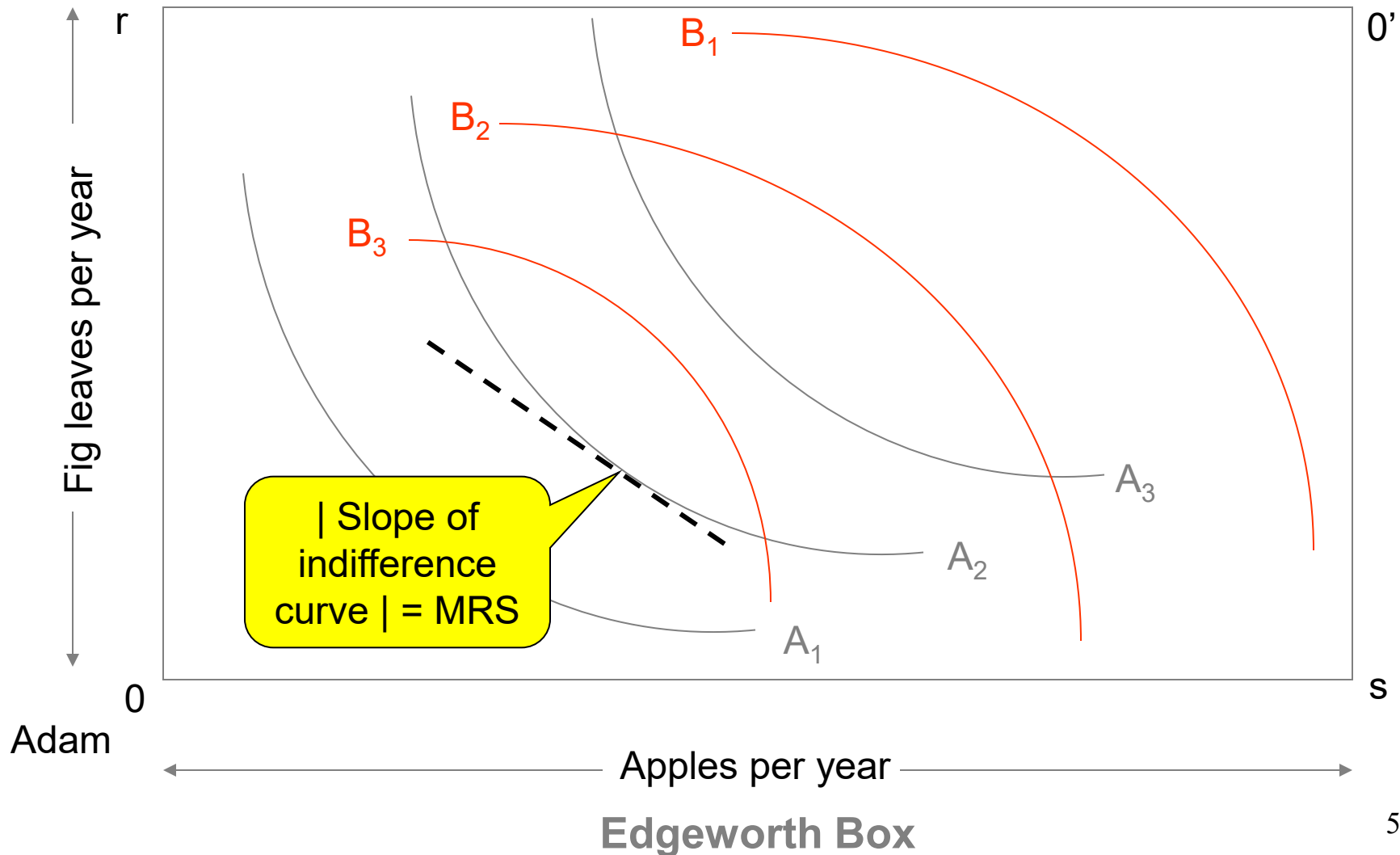
- Total amount of goods fixed
- Each point in EB gives a distribution of the goods

Recall about indifference curves

- Set of all consumption bundles that yield equal utility
- $|\text{Slope IC}| = \text{Marginal Rate of Substitution}$
- MRS gives ratio at which trading goods does not make individual better or worse off.
 - $\text{MRS} = \text{amount of good Y that compensates exactly for reduction in good X by 1}$
 - $\text{MRS} = \text{MU}^{\text{X}} / \text{MU}^{\text{Y}}$

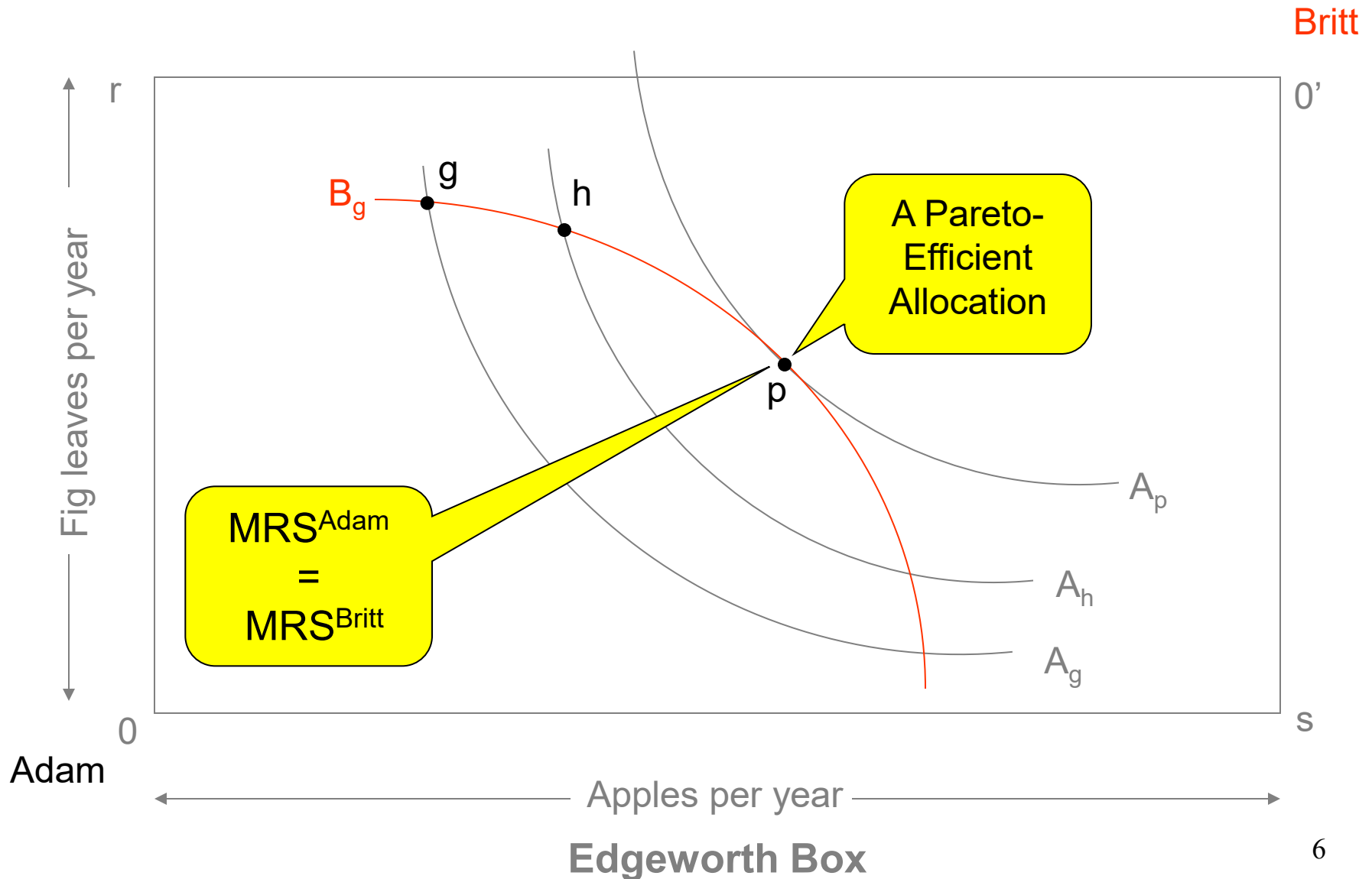
Indifference curves in Edgeworth Box

Britt



Efficiency-improving exchange:

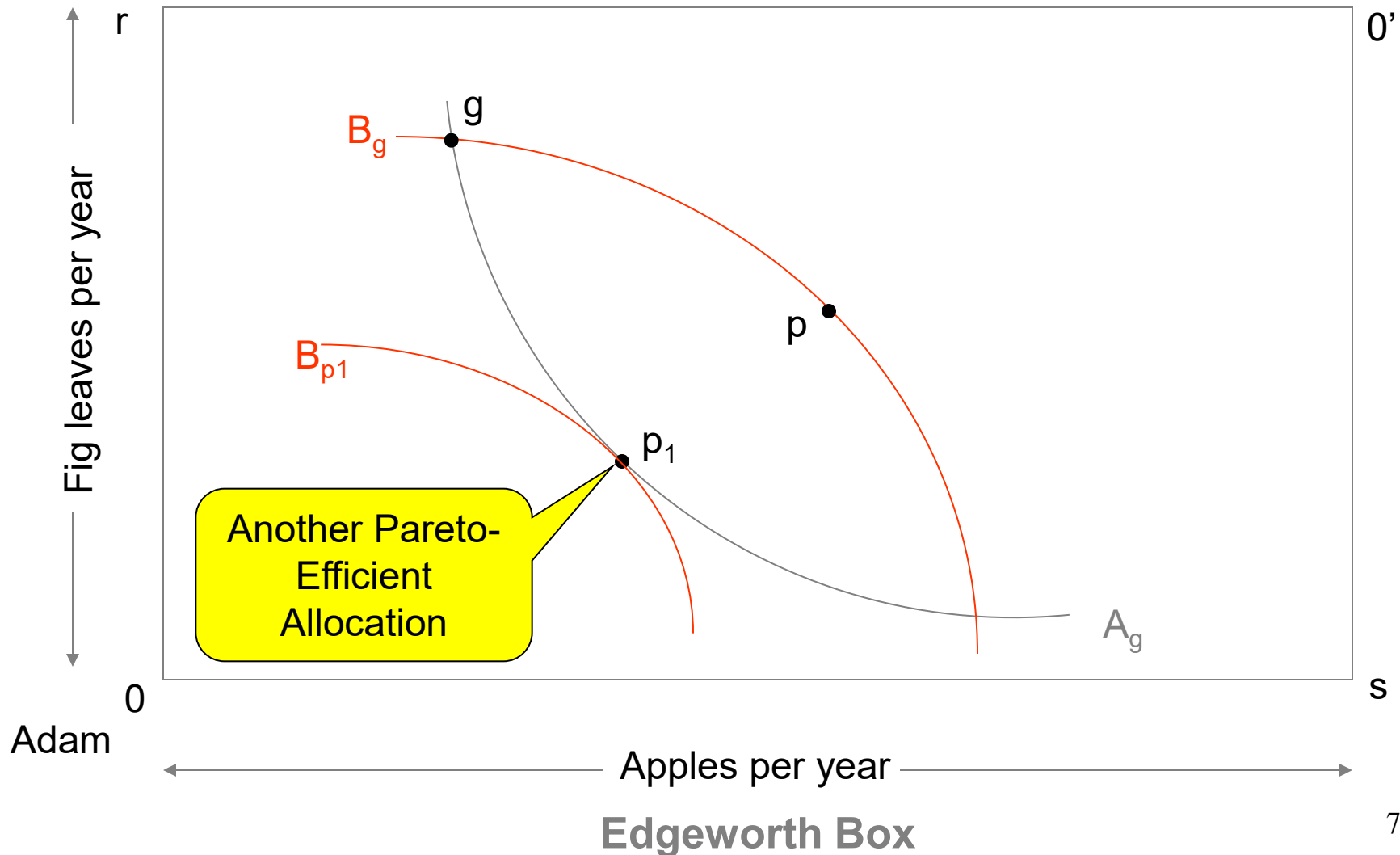
Making Adam better off without Britt becoming worse off



Efficiency-improving exchange:

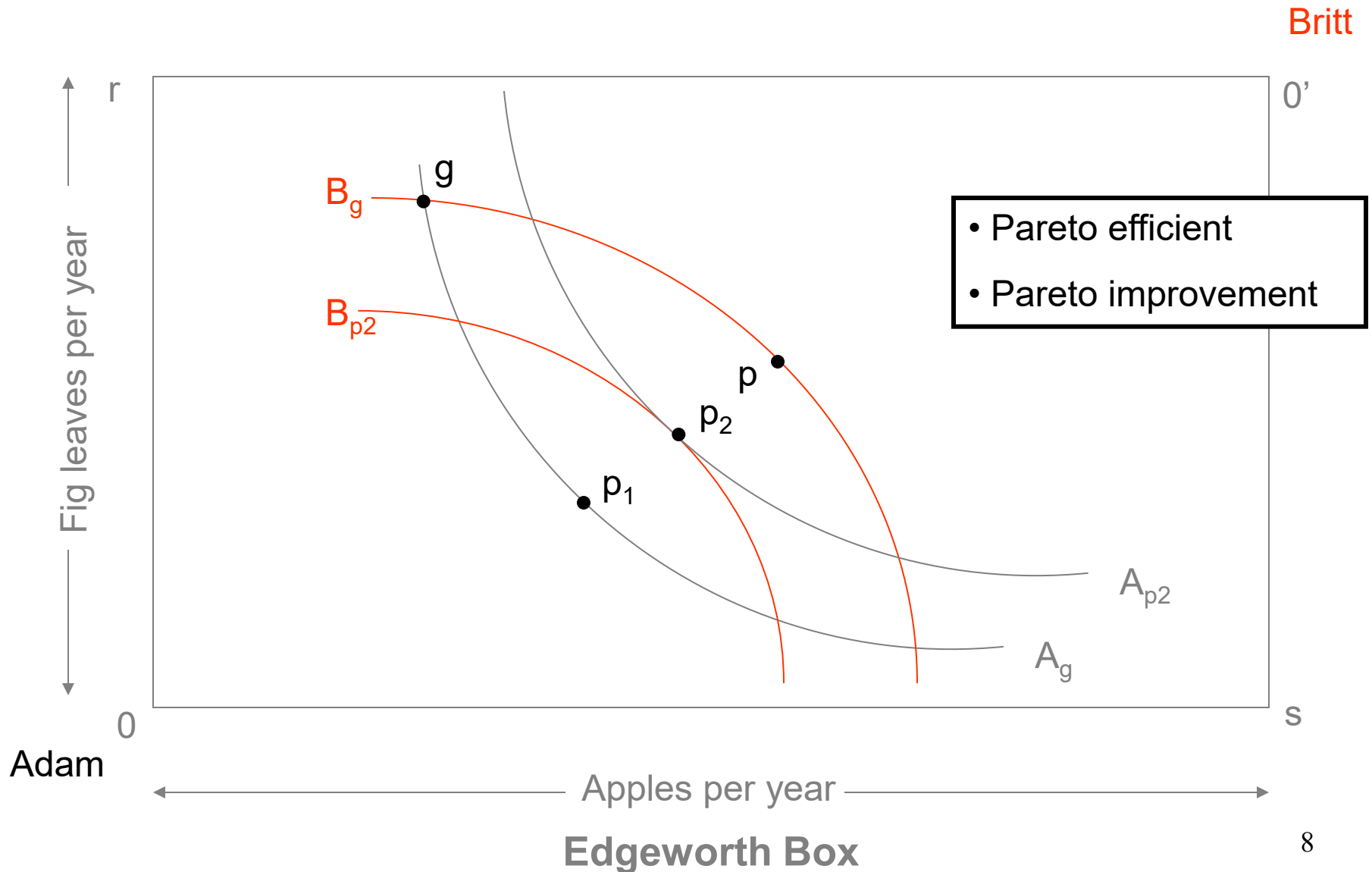
Making Britt better off without Adam becoming worse off

Britt

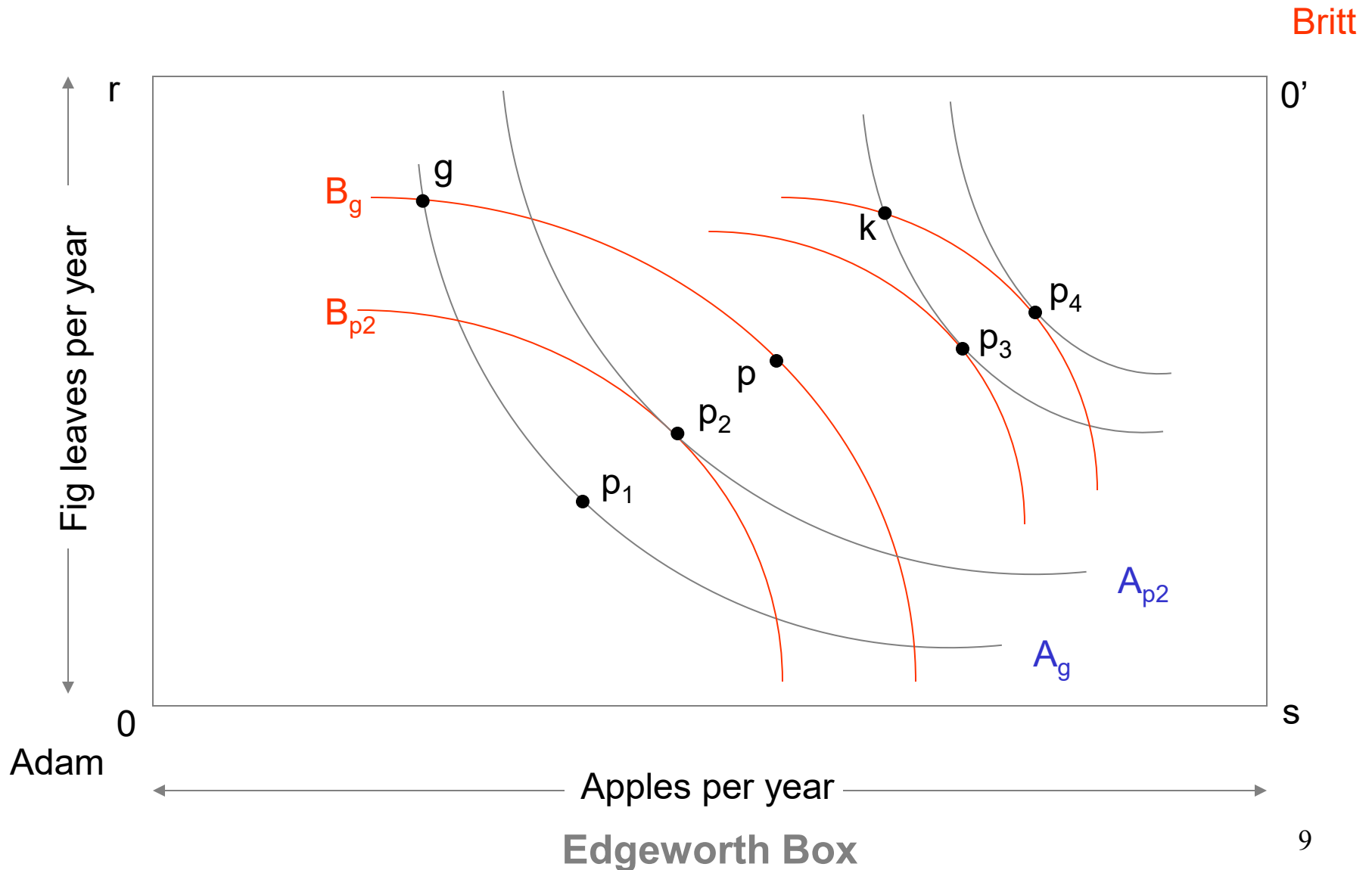


Efficiency-improving exchange:

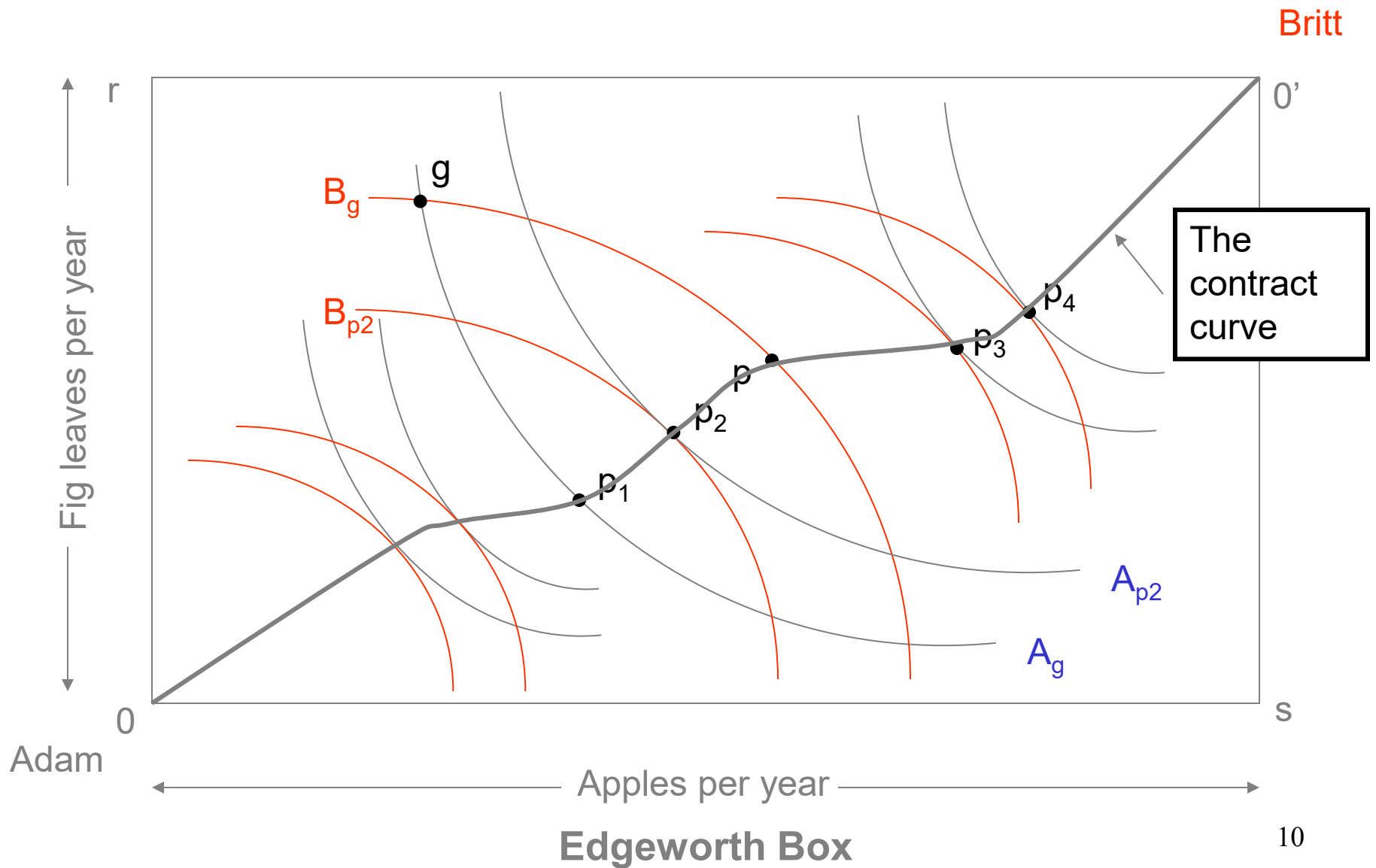
Making both Adam and Britt better off



Starting from different initial endowments



The Contract Curve



Perfect Competition

In exchange economy, total quantities of goods fixed.

In reality, economy can change allocation of inputs in production...

- Labour, (human) capital, land, etc.

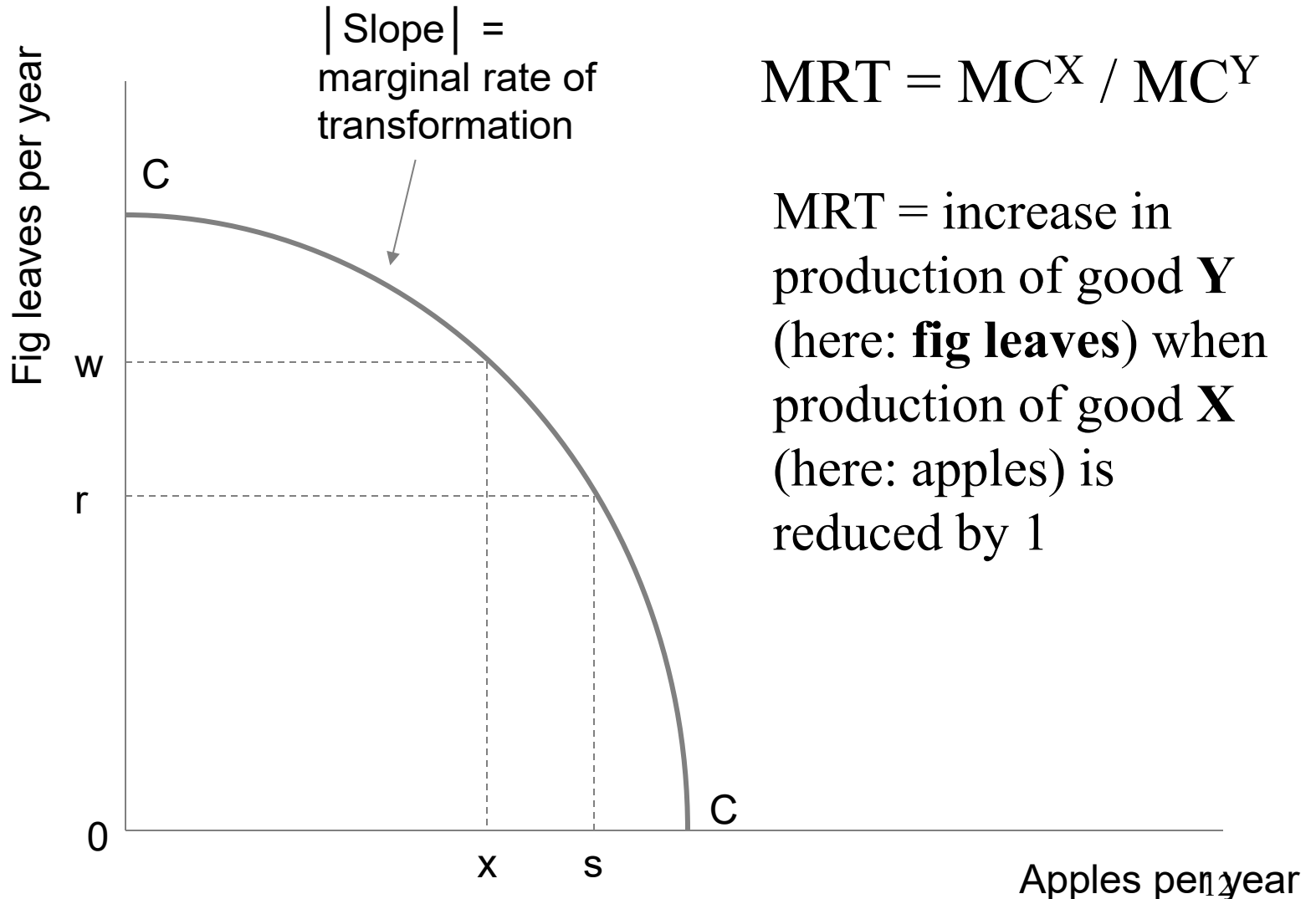
... to change ‘product mix’

→ Production Possibilities Frontier

Generally, more of good X implies less of good Y.

But how much? → Marginal Rate of Transformation

Production Possibilities Curve



Perfect Competition

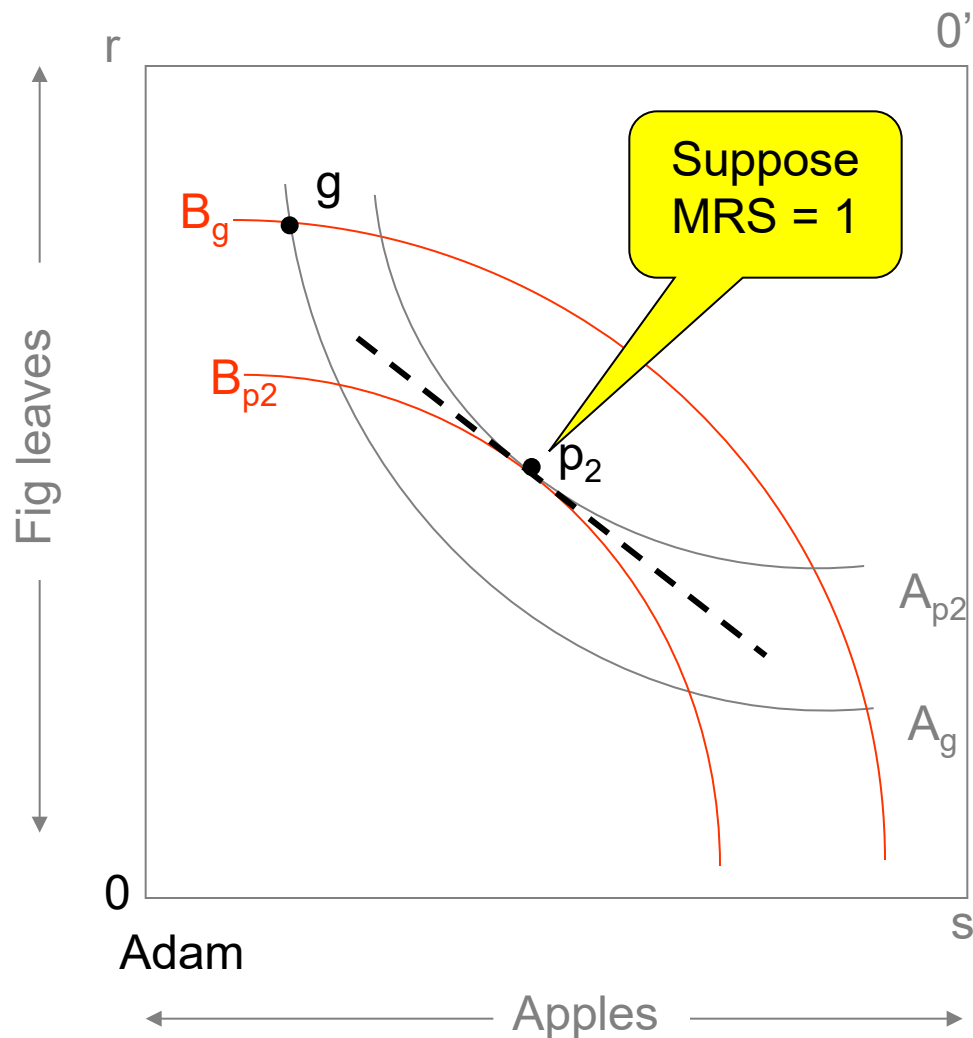
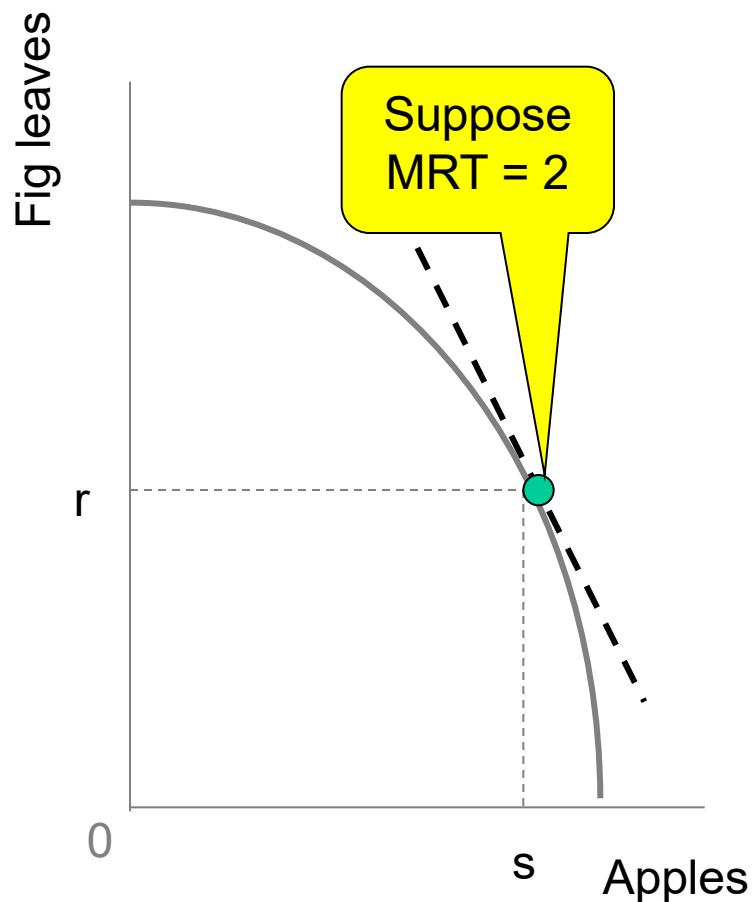
In pareto-efficient equilibrium:

$$MRS^A = MRS^B = MRT$$

- If $MRS \neq MRT$, different ‘product mix’ improves efficiency

Perfect Competition

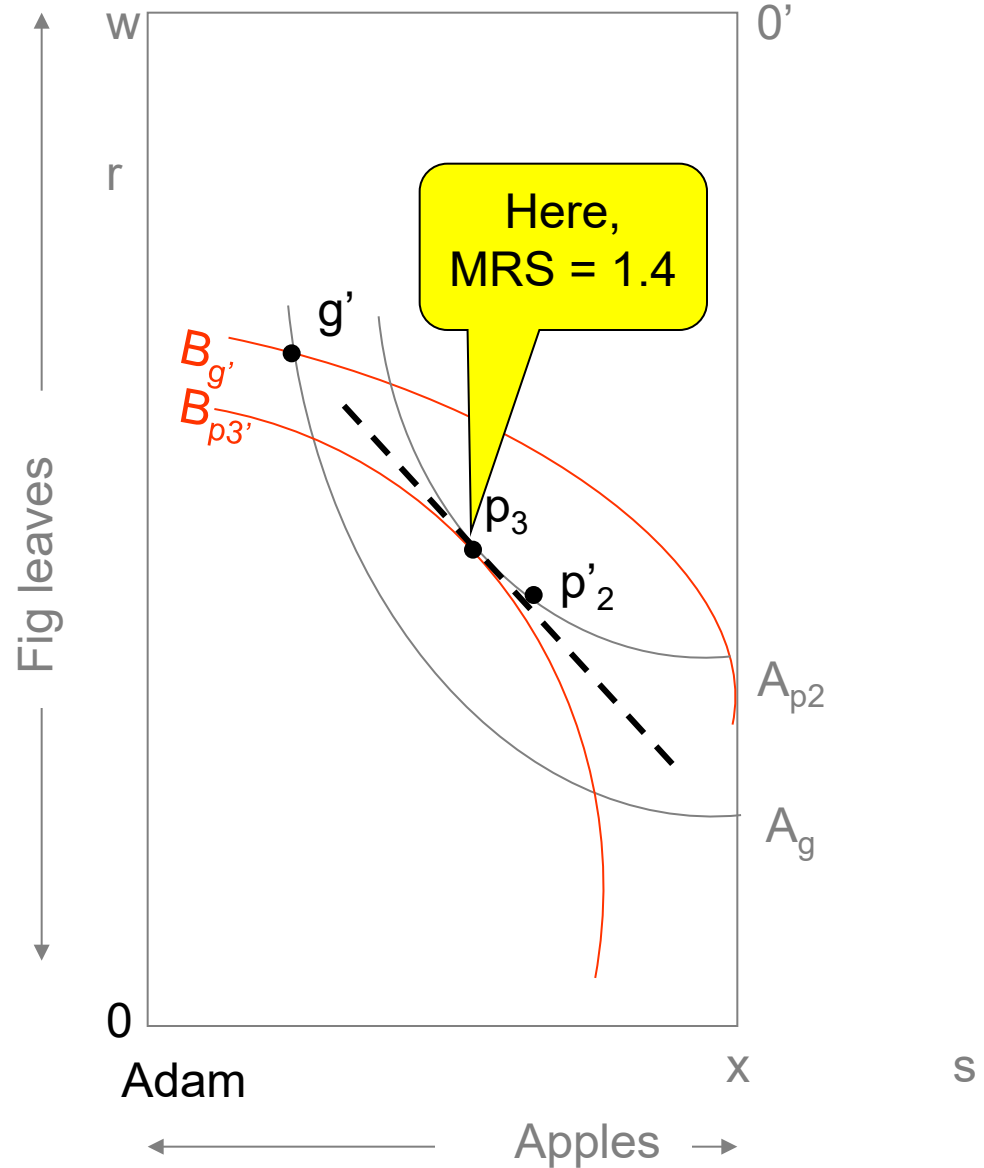
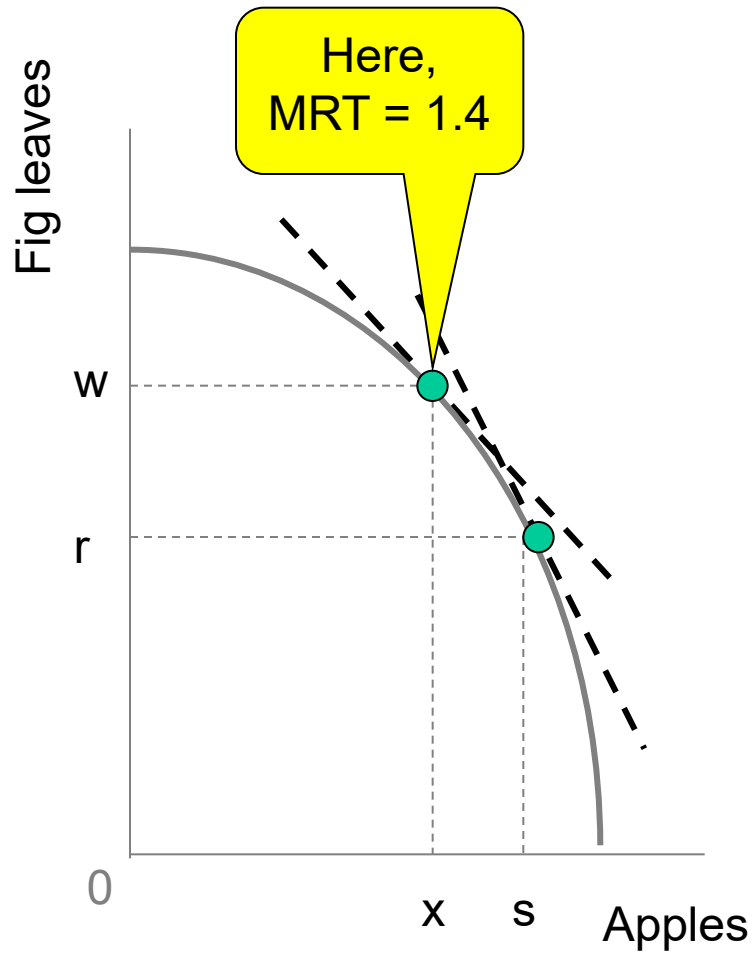
If $MRT \neq MRS$, we are not in equilibrium



Britt

Perfect Competition

Britt



Perfect Competition

Notes:

- Different product mix leads to different shape of Edgeworth Box
 - This implies that, comparing the two figures, a given consumption bundle for Adam (like point g) gives different consumption bundles to Britt.
- Adam has the same IC running through g and g' , but Britt has not!
- Same goes for p_2
- Therefore, we arrive in the second figure at p_3 rather than p_2

Perfect Competition

Exchanging goods may work in a two-by-two economy,

But how do we arrive at equilibrium with millions of people and goods? → Prices!

Given the assumption underlying Perfect Competition...

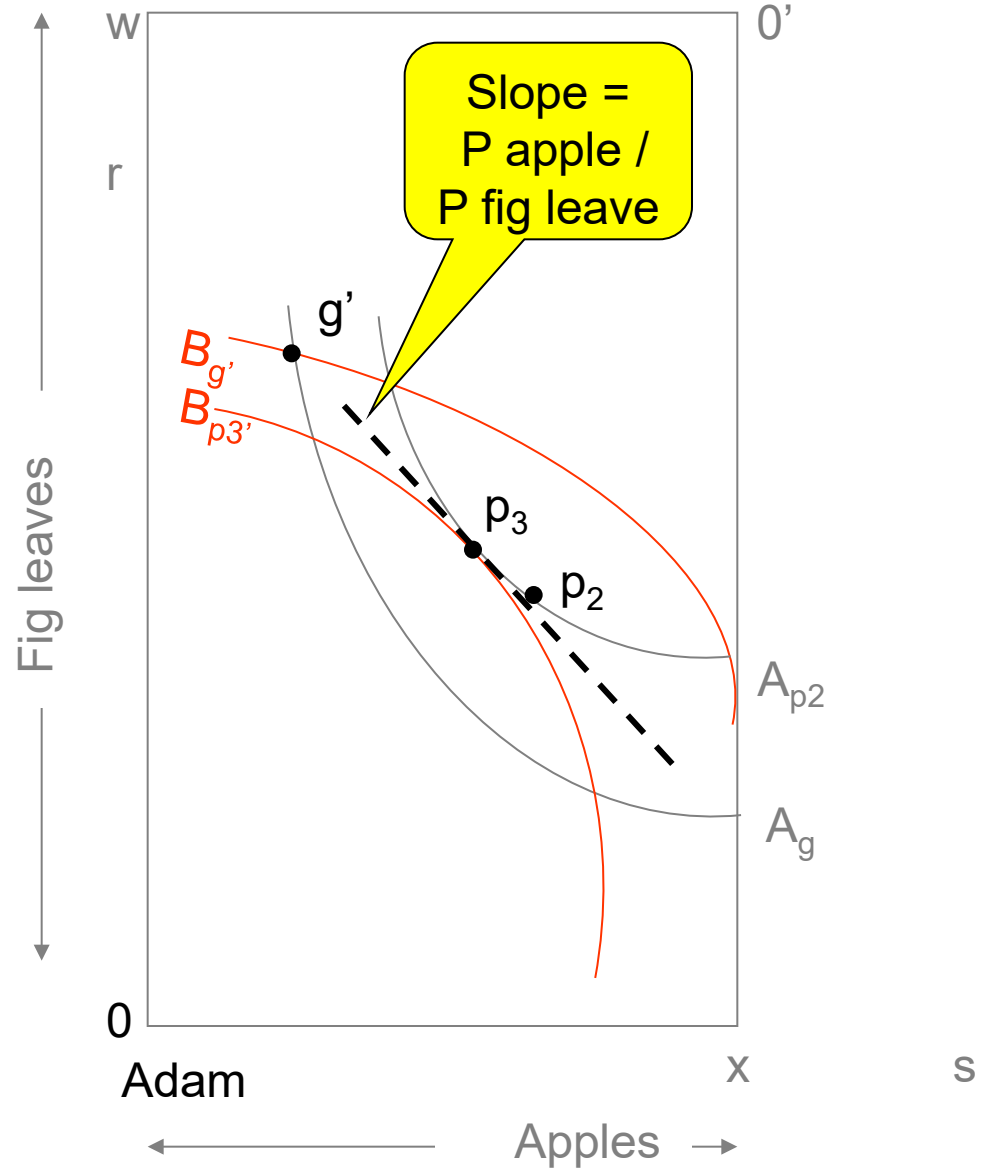
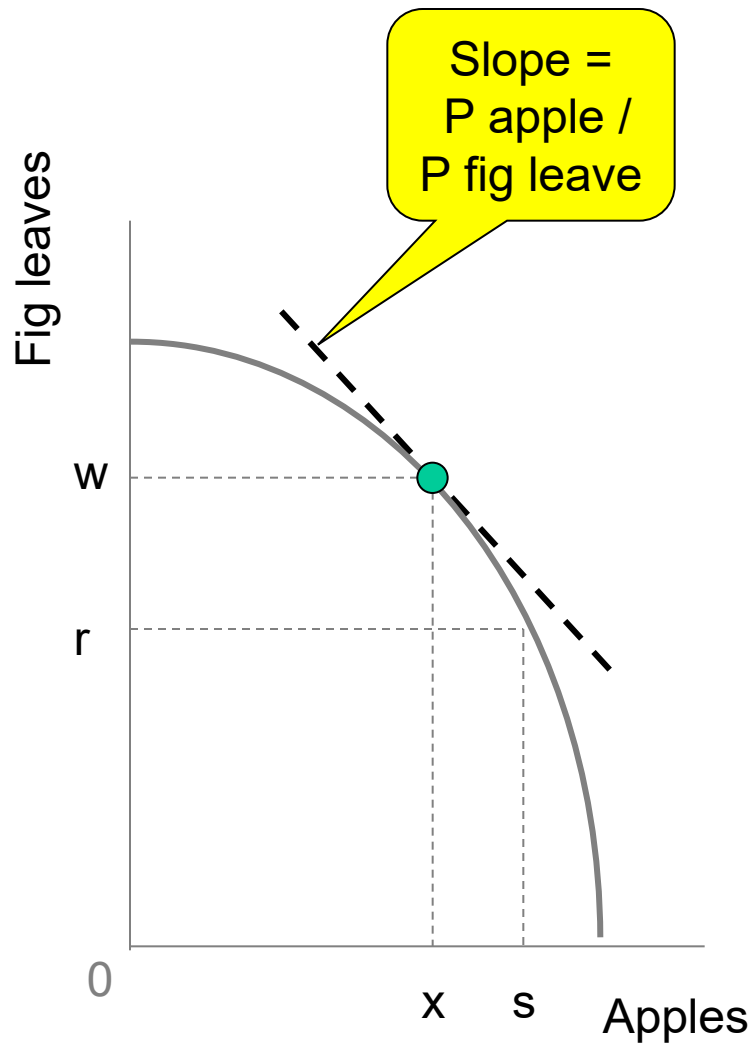
- Perfect information; many buyers and sellers; standardised products; ‘normal’ preferences;

... prices arise such that for everyone and for every combination of two goods X and Y, we have

$$MRS^A = MRS^B = \dots = MRT = \mathbf{P^X / P^Y}$$

Perfect Competition

Britt



Perfect Competition

Conclusion = First Fundamental Theorem of Welfare:

Perfectly competitive markets lead to Pareto-efficient outcomes.

But:

1. Is efficiency all we care about?
2. What if markets are not perfectly competitive?

Perfect Competition

- Is efficiency all we care about?

General answer: No, we also care about **distribution**

→ division of output across people

First FTW says nothing about distribution (or equity)

Second Fundamental Theorem of Welfare:

Any Pareto-efficient equilibrium can be attained by redistributing initial endowments

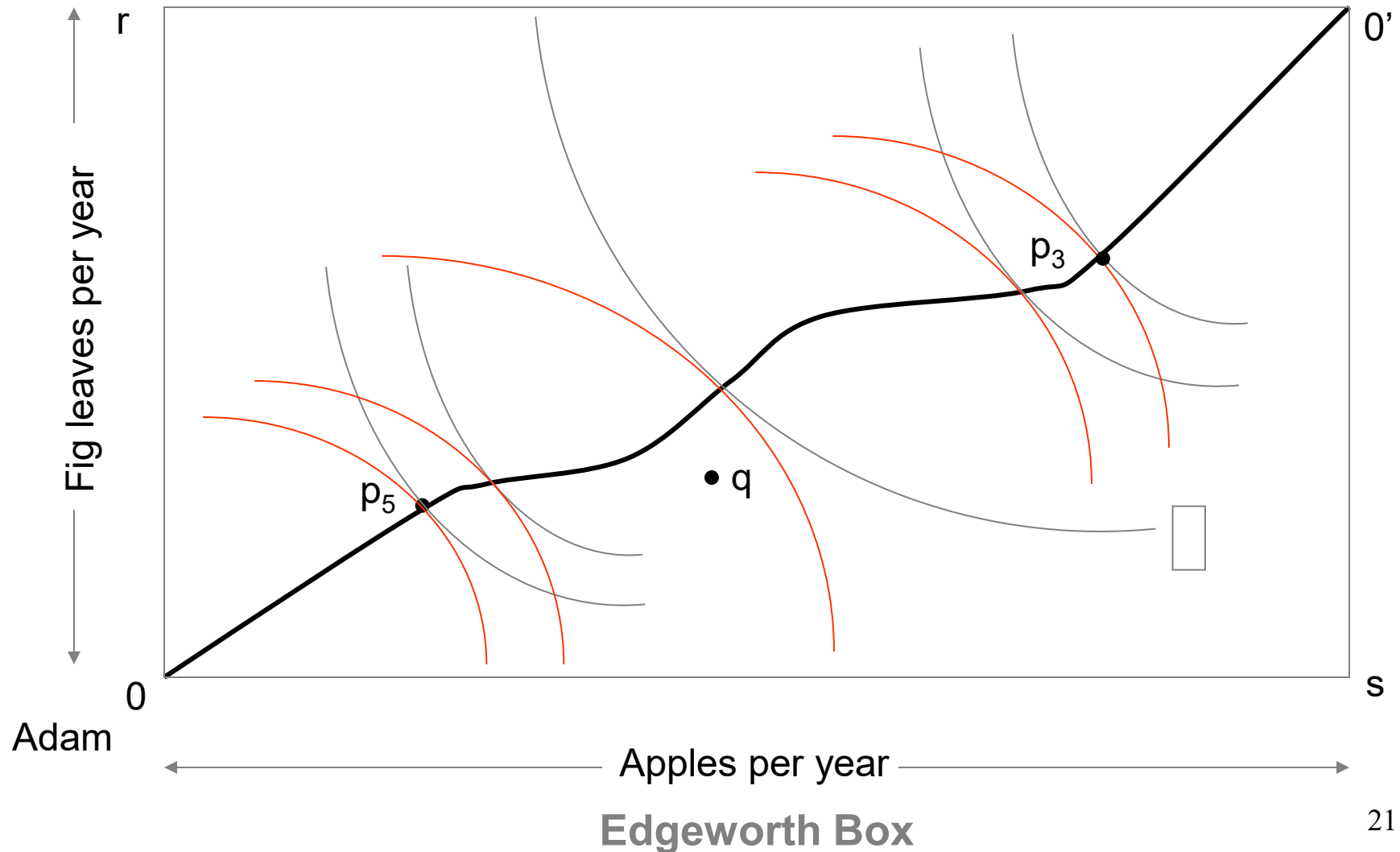
OK, but then: which outcome is ‘most desirable’?

→ How to choose among different (Pareto-efficient) equilibria?

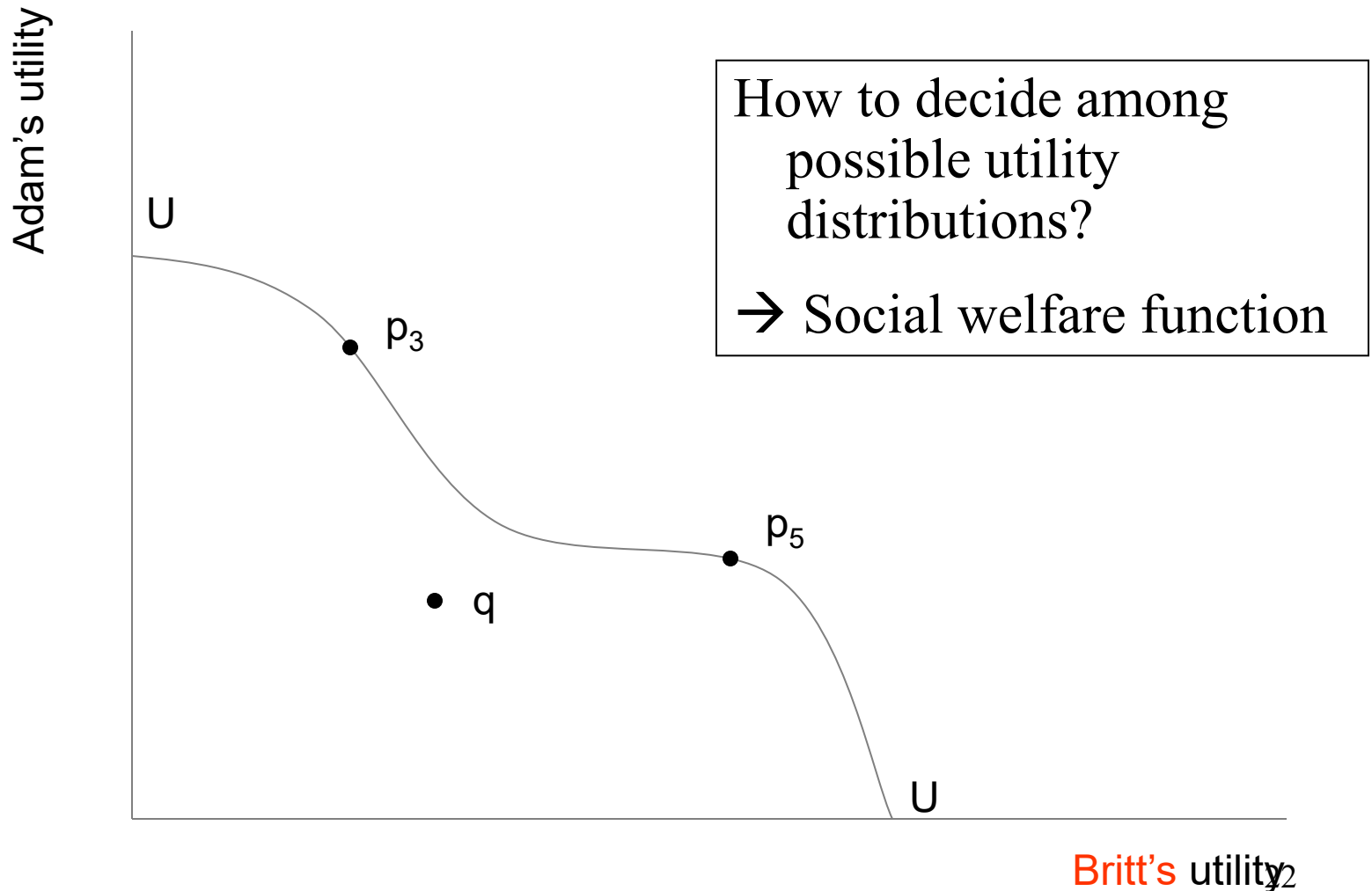
Efficiency versus Equity

Does 'society' care about which Pareto-eff. allocation arises?

Britt



Utility Possibilities Curve



Social Welfare Function

SW-function: $W = F(U^{Adam}, U^{Britt})$

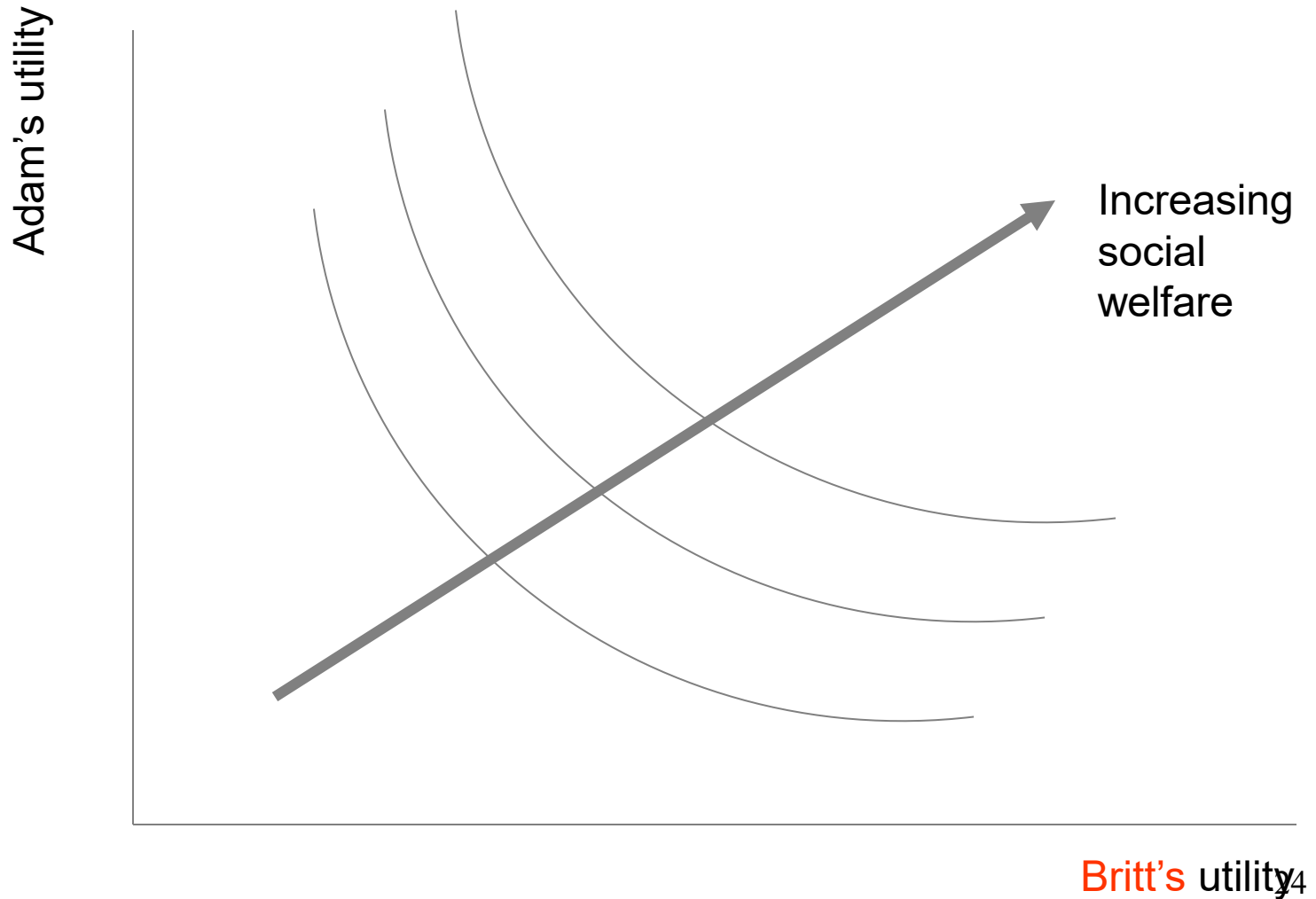
Comparing different utility distributions

- ‘Society’ prefers some distributions over others
- Provides ‘optimal’ distribution of utility

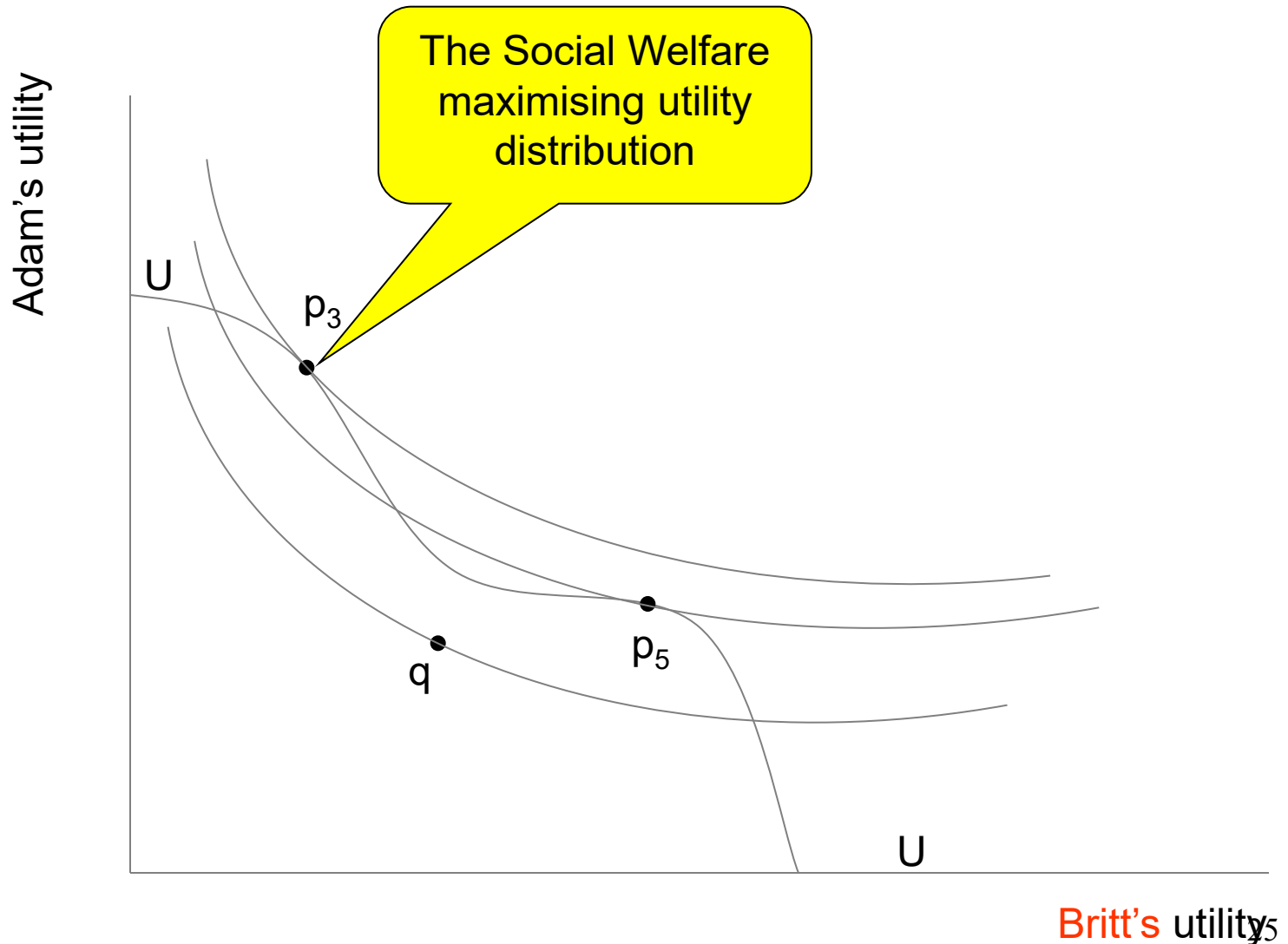
Specific examples:

- Utilitarian or Additive: $W = U^{Adam} + U^{Britt}$
- Rawlsian: $W = \min[U^{Adam}, U^{Britt}]$

Social Indifference Curves



Maximising Social Welfare



Social Welfare Function

Remarks:

- SW-function is a theoretical construct
- Useful for studying *normative* questions
- Recall from Philosophy: different views / approaches to ‘distributive justice’, including
 - Utilitarianism & prioritarianism
 - Libertarianism
 - Capabilities approach (Amartya Sen)
 - Marxism

Maximising Social Welfare

Second FTW: redistribute endowments such that market arrives at social optimum

→ The outcome that maximizes social welfare is also Pareto-efficient

In reality: redistribution is generally *income* redistribution → level of income can be affected by people

- Education; training; job choice; attitude towards work; etc.

Then: redistribution affects choices by changing relative prices
→ distortions, leading to lower efficiency

→ Trade-off between efficiency and distribution!

Main issue in weeks 4 – 6 of this course

Maximising Social Welfare

- What if markets are not perfectly competitive?

This reduces efficiency and, hence, shifts the UPC

It may also affect distribution

Government intervention may improve efficiency....

(weeks 1 to 3 of this course)

... and may affect distribution

Hence, as society cares about efficiency **and** distribution,
(proposals for) government intervention should be evaluated
on the effect on **both** criteria

(after the break)

Remainder of this lecture

- Public versus private provision of goods
 - Cost-benefit analysis
 - Efficiency and distribution
 - Valuing intangibles
- Public versus private production of goods

Public versus private provision

Public provision:

Government (partially) finances goods

Should the government (partially) pay for

- Defense
- Education
- Health care (or health insurance)
- Infrastructure and public transport
- Food and housing
- Textbooks in highschool
- Childcare
- Performing arts
- (research on) covid-vaccins
- Solar panels?

Public versus private provision

How to choose: cost-benefit analysis

In case of public goods, we have seen:

- Private provision may yield underprovision
- Public provision may yield overprovision

Other issues in this trade-off:

- Input cost and administrative cost
 - Private: better able to cater to individual tastes
 - Private: may lead to undesirable differences in consumption (e.g. in health care; education)
- ‘Commodity egalitarianism’

Public versus private provision

Evaluating proposals for government intervention:

1. What are the benefits: Does it improve either efficiency or redistribution?
 - What market failure is being addressed?
 - Why is the government able to do better?
 - Who benefits / loses?
2. At what cost (including tax distortions!)?
3. Are you sure the proposal does what it should do?
 - How will people respond?
 - Is it possible to circumvent, evade, or game the policy?

Cost-Benefit Analysis with Intangibles

General rule in investment decisions (R&G ch. 8):

Invest when discounted benefits $B >$ discounted cost C

- Examples: infrastructure; R&D

In case of public investment, several special issues

- Which discount factor to use?
 - Lower ‘social’ discount factor than market because society should value future generations?
- Distributional issues: who benefits?
- How to take non-monetary B’s and C’s into account?

Cost-Benefit Analysis with Intangibles

Example: Highway A4 between Schiedam & Delft

→ 7 km ‘missing’; track mostly through grassland
40 years of policy debate on this 7 km.

Cost-Benefit analysis:

- Benefit: Better traffic flow
- Cost: construction (€880million) + maintenance
- Cost: loss of nature

How to decide?

→ Somehow, loss of nature must be valued

Cost-Benefit Analysis with Intangibles

Similar issues arise in case of

- Parks
- Art in public places
- Clean versus littered streets
- Subjective feelings (safety; happiness; pride; social cohesion; etc.)

Some attempts at valuing intangibles:

- Ask people for their valuation of (hypothetical) plans
- Use (change in) housing prices → beautiful park or clean streets imply higher price (‘hedonic pricing’)

Special example: value of life

Life is priceless?

Not to economists, not to policy-makers (and certainly not to health insurers)

In all (public *and* private) decisions that involve safety, value of life plays a role, either explicitly or implicitly

- Safe vs risky job (→ wage differential)
- Roads / railways: crossing versus tunnels
- Regulations concerning car safety
- Transportation of hazardous substances
- Time-to-accident of ambulance / fire brigade
- Number of ICU-beds

Special example: value of life

Most visible in health care:

Quality-Adjusted Life Year (QALY)

QALY is a measure of the value of a life-year, adjusted for disabilities

QALY used to evaluate new medicines / treatments

→ If new medicine slightly better but much more costly, then it is not approved.

Examples:

- Medication for Pompe's disease or cystic fibrosis (taaislijmziekte)
- Breast-screening for women above 75.

Public versus private production

Public production:

Government not only finances goods, but also produces them → owns production factors

Examples:

- Army; police
- National Health System in UK
- Before 1980's: many big firms (KPN; BP)
- Until recently: public utilities (energy; water; etc.)

Public versus private production

Even in case of public provision, private firms might produce goods / services

Many of these tasks could (at least in theory) be performed by private companies (possibly not-for-profit):

- Police
- Prisons (there are private prisons in the US)
- Museums
- Harbor of Rotterdam

Public versus private production

Big issue, both in public debate and in scientific literature.

Two main arguments used in this debate

- Market environment / competition makes private producers more efficient → cheaper production
 - Private producers might cut back on (unobservable) quality to increase profit
- Less of a problem when there are multiple firms and a long time-horizon: reputation-building

Concluding remarks

Public vs private provision / production is hotly debated topic

This course provides no clear-cut answers, but does provide a framework for looking at these issues

To assess any proposal for government intervention, ask:

- Does it improve efficiency? (or: which market failure is being addressed?)
- Does it have favourable redistributive effects?
- Can it be implemented at reasonable cost (including tax distortions)?

Note: answers may not be readily available

Next week: externalities

Issues ranging from:

- The ‘real’ cause of global warming

to:

- The (too?) low level of your tuition fee

and to:

- The importance of establishing and enforcing property rights
- The main argument for government intervention in a pandemic